

ENGLISH FOR WORK / SEPTEMBER 2026

Engineering English



Learner workbook

Work with realistic cases, develop your own response, and use the answer section after you have tried the tasks.

PRACTICE THAT TRANSFERS TO WORK

Read a realistic case. Study complete conversations. Find the right language, speak, get feedback, and try again.

Eight lessons | Intermediate to advanced | Independent or classroom study

For mechanical, electrical, civil, systems, industrial, test, quality, manufacturing, and field engineers, plus engineering managers and technical project leads.

These cases practice communication. Actual equipment, construction, and safety work follows approved procedures and authorized specialist decisions.

All scenarios, figures, and model responses are fictional teaching material. Use invented details in practice.

[Open this course on English Ladder](#)

Your course at a glance

Use the lesson numbers to match this document with the website and the other guides.

01 Requirements and Constraints

Clarify missing information before responding.

02 Design Reviews and Technical Pushback

State a concern respectfully and propose a practical route forward.

03 Failure Modes and Reliability

Separate observations, assumptions, and conclusions.

04 Testing, Validation, and Data Interpretation

Explain a useful distinction in plain English.

05 Manufacturability and Cost Engineering

Compare options using the same criteria and state the tradeoff.

06 Safety Factors and Compliance

Make a request with a clear action, purpose, and realistic timing.

07 Field Issues and Customer Communication

Acknowledge a communication problem and give a corrected message.

08 Systems Integration and Interface Control

Give a handoff that the receiving person can confirm.

Choose your pace

Quick practice: spend 15 minutes reading one conversation and rehearsing one scenario. Full lesson: allow 45-60 minutes for language work, three conversations, two speaking scenarios, and feedback.

Level support

B1 learners can use the frames and a partner. B2 learners can try a response before reading the model. C1 learners can add the harder follow-up and reduce preparation time. These are teaching suggestions, not certified CEFR level ratings.

After practicing: open the AI prompt workshop to choose a vocabulary, grammar, role-play, or new-dialogue activity.

LESSON 01 / READ AND NOTICE

Requirements and Constraints

Clarify missing information before responding.

SITUATION

A customer asks for a "lightweight and durable" enclosure without specifying weight, operating conditions, or required life.

1. Understand the situation

Identify two confirmed facts, one missing detail, and the person who needs your response. Do not fill gaps with invented facts.

2. Find the words

requirement - A condition or capability that must be met for a stated purpose.

constraint - A limit that restricts the available choices or amount of work possible.

tolerance - The permitted variation from a specified value or dimension.

tradeoff - A choice that improves one objective while giving up something on another.

3. Notice the language

Name the exact word, fact, or requirement you need clarified. Ask one focused question at a time. Repeat your understanding and give the other person a chance to correct it.

When you say ..., do you mean ...?

Could you clarify which ...?

So, my understanding is Is that right?

Compare this wording: Could you explain what does this mean?

LESSON 01 / PRACTICE AND PRODUCE

Ask a precise question

4. Check the language

1. Which indirect question has the usual word order?
 - A. Could you confirm when the review starts?
 - B. Could you confirm when does the review start?
 - C. Could you confirm when starts the review?
2. Someone says the report is 'ready.' Which question best clarifies its approval status?
 - A. Can you send the report to me this afternoon?
 - B. Who will present the report at the meeting?
 - C. Has the report been approved, or is it ready for review?

05 • Conversations

The next three pages contain original fictional conversations for this lesson. Each numbered line is one speaking turn. Read the setting before assigning roles.

LESSON 01 / 05 • CONVERSATIONS / 1 OF 3

What does lightweight mean for this enclosure?

Fictional requirements meeting: a customer requests a lightweight, durable enclosure without numerical limits or operating conditions.

01 • Mechanical engineer: When you say lightweight, do you mean easy to carry or below a specified mass?

02 • Customer engineer: Easy to carry during inspections. We haven't agreed a maximum mass for the complete unit.

03 • Mechanical engineer: Does that include the battery and handle, or only the enclosure itself?

04 • Customer engineer: The complete unit matters to users. My original request wasn't clear about that boundary.

05 • Mechanical engineer: Thanks. What operating conditions and required life should inform the durability requirement?

06 • Customer engineer: Those are still being collected. I cannot give you a verified temperature range today.

07 • Mechanical engineer: Then let's avoid choosing material from an assumed range or treating lightweight as testable yet.

08 • Customer engineer: Could you identify the missing constraints so I can ask the product team focused questions?

09 • Mechanical engineer: Yes: total mass limit, handling conditions, operating environment, and expected service life.

10 • Customer engineer: I'll clarify those before we discuss the weight and durability tradeoff in detail.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 01 / 05 • CONVERSATIONS / 2 OF 3

A close fit without a defined clearance

Fictional fixture discussion: a drawing calls for a close fit around a removable insert but specifies no clearance.

01 • Design engineer: The drawing says close fit. What must the operator be able to do with the insert?

02 • Manufacturing engineer: Remove it by hand for cleaning, without using a lever or striking the fixture.

03 • Design engineer: Then close fit describes a preference, but we still need a measurable clearance requirement.

04 • Manufacturing engineer: Can we just use the nominal insert dimension and allow no visible movement?

05 • Design engineer: That ignores tolerance, meaning the permitted dimensional variation in the parts we actually receive.

06 • Manufacturing engineer: You're right. We also haven't confirmed the insert supplier's dimensional range for this batch.

07 • Design engineer: Please obtain that range before we set a mating dimension that could prevent removal.

08 • Manufacturing engineer: Should we include the hand-removal requirement in the functional review rather than only the drawing?

09 • Design engineer: Yes. We need both measurable dimensions and a check of the intended handling task.

10 • Manufacturing engineer: I'll bring the supplier information and flag clearance as unresolved until we review it.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 01 / 05 • CONVERSATIONS / 3 OF 3

Quiet at which operating point?

Fictional fan-cover project: a product brief asks for quiet operation, but speed, measurement distance, and acoustic target are unspecified.

01 • Product engineer: The cover must support quiet operation. Could you clarify the condition behind that requirement?

02 • Product manager: Customers notice noise at their desks, but we haven't specified fan speed or measurement distance.

03 • Product engineer: Those details matter because two tests under different conditions may not be comparable.

04 • Product manager: Could we adopt the competitor's advertised number as our acceptance target for now?

05 • Product engineer: Only after understanding its test conditions. Otherwise the comparison could give us a misleading constraint.

06 • Product manager: Then what can we agree today without pretending we already have a valid acoustic specification?

07 • Product engineer: Agree the intended desk use and assign someone to define a representative operating condition.

08 • Product manager: I'll confirm that use case. The noise target itself will remain an open requirement.

09 • Product engineer: Good. Once we have it, we can assess the airflow and acoustic tradeoff consistently.

10 • Product manager: Bring the missing measurement questions to our next review so we can resolve them explicitly.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 01 / 06 • SAY IT

Two scenarios to practice

Use only the supplied facts. Prepare for two minutes, speak for one minute, then respond to a follow-up. Switch roles and repeat. For solo study, speak both roles aloud.

Scenario 1 | Requirements and Constraints

A customer asks for a "lightweight and durable" enclosure without specifying weight, operating conditions, or required life.

Prepare for two minutes. Speak for 45-60 seconds using the case facts, then respond to your partner's question. Switch roles and repeat without reading the model response.

Partner: You need to know exactly what information is missing. Give one answer only after your partner asks a focused question; then ask them to confirm their understanding.

Second round: The other person uses an ambiguous word such as 'ready', 'approved', or 'urgent'. Clarify it before continuing.

Model for scenario 1: Could you define the target weight, operating conditions, and expected life? We need measurable requirements to discuss the tradeoffs and evaluate a design consistently.

Scenario 2 | A cable that must reach

A fictional test station needs a replacement cable described as long enough. The equipment locations are known, but routing clearance, connector orientation, and required slack have not been checked. Clarify the requirement before ordering.

Role A: Design engineer: ask focused questions about the routing constraint.

Role B: Test technician: explain the station layout without inventing an unmeasured cable length.

Possible opening: Before choosing a cable length, could we confirm the route and the slack needed for maintenance?

Success checks

- Separate direct distance from routed length.
- Identify connector and maintenance constraints.
- Agree a measurement check before selecting a length.

Add a complication: The technician suggests using the straight-line distance as the complete requirement.

Check: facts accurate | meaning clear | tone appropriate | next action or question explicit

Continue scenario 1 with an AI partner.

LESSON 02 / READ AND NOTICE

Design Reviews and Technical Pushback

State a concern respectfully and propose a practical route forward.

SITUATION

A design review is moving toward approval while one interface assumption remains undocumented.

1. Understand the situation

Identify two confirmed facts, one missing detail, and the person who needs your response. Do not fill gaps with invented facts.

2. Find the words

design review - A structured evaluation of a design at a defined stage.

design rationale - The reasons and evidence supporting a design choice.

risk - Uncertainty that may affect objectives or expose people or assets to harm or loss.

verification - Checking whether specified requirements have been met.

3. Notice the language

Acknowledge the goal without pretending agreement. Name the specific problem and its implication, then ask a focused question or propose a next step. Be direct when a safety or ethical boundary requires it.

I understand the goal. My concern is

That statement goes beyond

Could we ... before confirming ...?

Compare this wording: That idea is ridiculous.

LESSON 02 / PRACTICE AND PRODUCE

Disagree with a useful reason

4. Check the language

1. Which response identifies a specific assumption and evidence against it?
 - A. The proposal appears optimistic, and I would prefer a slower approach.
 - B. I would like another discussion before we proceed with the proposal.
 - C. The proposal assumes the review is complete; the record shows it is still open.
2. Which follow-up makes disagreement actionable?
 - A. Could we ask for general feedback before sending the next update?
 - B. Could we resolve the missing approval before confirming the plan?
 - C. Could we confirm the plan and discuss the missing approval later?

05 • Conversations

The next three pages contain original fictional conversations for this lesson. Each numbered line is one speaking turn. Read the setting before assigning roles.

LESSON 02 / 05 • CONVERSATIONS / 1 OF 3

Approval with an undocumented interface assumption

Fictional design review using the lesson case: approval is being discussed while an interface assumption remains undocumented.

01 • Review chair: The design looks ready. Can we record approval and address minor documentation later?

02 • Systems engineer: My concern is the undocumented interface assumption. We haven't established whether it is minor.

03 • Review chair: Which risk does that create, beyond making the design rationale harder to read?

04 • Systems engineer: If the connected subsystem behaves differently, the verification plan may test the wrong conditions.

05 • Review chair: Are you saying the design is wrong, or that its basis needs confirmation?

06 • Systems engineer: Its basis needs confirmation. I don't have evidence that the design itself is wrong.

07 • Review chair: What practical step would resolve the concern without repeating the whole design review?

08 • Systems engineer: Ask the interface owner to confirm the assumption and identify any affected verification cases.

09 • Review chair: I'll keep that item open and separate it from the parts already reviewed.

10 • Systems engineer: Thank you. Then the decision record will reflect the actual uncertainty rather than implied agreement.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 02 / 05 • CONVERSATIONS / 2 OF 3

The service opening has no tool-clearance check

Fictional product review: a compact cover design is preferred, but maintenance-tool access has not been checked.

01 • Industrial designer: The smaller cover meets our appearance goal. Can we freeze this geometry today?

02 • Service engineer: I support the compact layout. My concern is that we haven't checked maintenance-tool clearance.

03 • Industrial designer: The fastener is visible in the model. Isn't visibility enough to establish access?

04 • Service engineer: Not necessarily. Seeing a fastener and fitting the intended tool are different checks.

05 • Industrial designer: Do you have a confirmed interference, or is this still a design-review question?

06 • Service engineer: Still a question. I want verification before we claim the service opening is adequate.

07 • Industrial designer: Could we evaluate the tool envelope using the existing model before making another prototype?

08 • Service engineer: That is a useful first step, followed by whatever physical check the review identifies.

09 • Industrial designer: I'll retain the current geometry as proposed and add the access rationale to the review.

10 • Service engineer: I'll provide the intended tool information so we can close the question with evidence.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 02 / 05 • CONVERSATIONS / 3 OF 3

Removing a test because one prototype passed

Fictional review: one prototype survived a handling demonstration, and a lead proposes removing a planned repeated-handling test.

01 • Project engineer: The prototype survived the demonstration. Could we remove the repeated-handling test to save time?

02 • Reliability engineer: I understand the schedule pressure, but one demonstration doesn't address repeated handling under the test plan.

03 • Project engineer: Would keeping the test imply that the demonstration had no value for the design review?

04 • Reliability engineer: No. It offers evidence for what happened once, not for every condition we planned to verify.

05 • Project engineer: Then can we use that evidence to narrow the test rather than cancel it outright?

06 • Reliability engineer: We can review that proposal against the requirement and document the design rationale for any change.

07 • Project engineer: I don't have authority to alter the agreed verification scope on my own.

08 • Reliability engineer: Let's bring the proposal to the review owner with the evidence and remaining risk clearly stated.

09 • Project engineer: I'll stop describing the test as redundant until that review is complete.

10 • Reliability engineer: Good. We can discuss efficiency without treating an unreviewed reduction as already approved.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 02 / 06 • SAY IT

Two scenarios to practice

Use only the supplied facts. Prepare for two minutes, speak for one minute, then respond to a follow-up. Switch roles and repeat. For solo study, speak both roles aloud.

Scenario 1 | Design Reviews and Technical Pushback

A design review is moving toward approval while one interface assumption remains undocumented.

Prepare for two minutes. Speak for 45-60 seconds using the case facts, then respond to your partner's question. Switch roles and repeat without reading the model response.

Partner: You want to move quickly and initially overlook a relevant condition. Ask why it matters, then consider a practical route forward that respects the real constraint.

Second round: The other person repeats the request more firmly. Restate the specific concern calmly and explain the appropriate next route.

Model for scenario 1: I support moving the review forward, but this interface assumption needs to be explicit. Could we record it and identify the verification needed before treating that part of the design as settled?

Scenario 2 | A new adhesive without reviewed evidence

A fictional assembly design uses an adhesive specified in the current drawing. A purchasing colleague proposes a substitute with a shorter lead time. Compatibility evidence is unavailable. Raise a technical concern and agree a review route, not an adhesive recommendation.

Role A: Design reviewer: explain why shorter lead time does not establish suitability.

Role B: Purchasing engineer: explain the delivery concern and ask for the evidence needed.

Possible opening: The shorter lead time helps, but we have not verified that the substitute meets this assembly's requirements.

Success checks

- Acknowledge the delivery goal.
- Distinguish a proposal from verified suitability.
- Agree who will review the substitution evidence.

Add a complication: The colleague says identical packaging descriptions should be enough.

Check: facts accurate | meaning clear | tone appropriate | next action or question explicit

Continue scenario 1 with an AI partner.

LESSON 03 / READ AND NOTICE

Failure Modes and Reliability

Separate observations, assumptions, and conclusions.

SITUATION

A component fails during one test. The team has not reproduced the failure or checked the test setup.

1. Understand the situation

Identify two confirmed facts, one missing detail, and the person who needs your response. Do not fill gaps with invented facts.

2. Find the words

failure mode - A specific way in which a component, process, or system can fail.

FMEA - Failure mode and effects analysis, a structured method for identifying how a process or product can fail and prioritizing controls.

reliability - The ability to perform an intended function consistently under stated conditions over time.

duty cycle - The pattern of operating time and conditions used to describe equipment use.

3. Notice the language

Say what the evidence shows, identify what is still unknown, and limit the conclusion to that evidence. Words such as 'may', 'suggests', and 'in this sample' are useful when the uncertainty is real. Do not soften an urgent, verified safety concern.

The available evidence shows

This may indicate ..., but

We have not yet established whether

Compare this wording: The pilot proves this will work everywhere.

LESSON 03 / PRACTICE AND PRODUCE

Match certainty to evidence

4. Check the language

1. Which sentence preserves uncertainty about an unconfirmed cause?
 - A. The change definitely caused the delay.
 - B. The change may have contributed to the delay.
 - C. The change must have caused the delay.
2. A result comes from a small pilot in one location. Which phrase accurately limits the claim?
 - A. In this pilot, the observed result was ...
 - B. In similar locations, we can now expect the same result ...
 - C. Across the service, the observed result was ...

05 • Conversations

The next three pages contain original fictional conversations for this lesson. Each numbered line is one speaking turn. Read the setting before assigning roles.

LESSON 03 / 05 • CONVERSATIONS / 1 OF 3

One failure, several open explanations

Fictional reliability meeting using the lesson case: a component failed once, and neither reproduction nor test-setup checks are complete.

01 • Test engineer: The component stopped during one test. I recorded the observed failure mode and test sequence.

02 • Reliability lead: Can we conclude the component design is unreliable, or do we need more investigation?

03 • Test engineer: We need more. The failure hasn't been reproduced, and the test setup hasn't been checked.

04 • Reliability lead: What should enter the failure mode and effects analysis, or FMEA, at this stage?

05 • Test engineer: The observation and possible explanations, clearly distinguished from a verified cause or failure frequency.

06 • Reliability lead: Do we know whether the test duty cycle matched the intended pattern of use?

07 • Test engineer: Not yet. I'll compare the recorded sequence with the specified on-and-off operating pattern.

08 • Reliability lead: Please preserve the original evidence so the setup review doesn't erase what actually happened.

09 • Test engineer: I'll retain the records and propose controlled follow-up checks through our test process.

10 • Reliability lead: Then we'll discuss reliability implications after reviewing the setup and reproduction evidence.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 03 / 05 • CONVERSATIONS / 2 OF 3

Wear after a different duty cycle

Fictional fixture trial: two samples show different wear; one ran continuously and the other ran intermittently.

01 • Production engineer: Sample A has more wear than sample B. Does that prove its material is inferior?

02 • Reliability engineer: Not from this comparison alone. Sample A ran continuously, while sample B ran intermittently.

03 • Production engineer: So the duty cycle differs, even though both samples spent the same week in the laboratory.

04 • Reliability engineer: Exactly. Calendar time doesn't tell us how much operating exposure each sample received.

05 • Production engineer: Should the FMEA list the material as the root cause of the observed wear?

06 • Reliability engineer: Not yet. Record wear as the observed failure mode and keep material differences as a hypothesis.

07 • Production engineer: What information would make the comparison more useful before we plan another test?

08 • Reliability engineer: Operating hours, loads, and the documented duty cycles would help us understand the exposure.

09 • Production engineer: I'll collect those records and avoid ranking materials from the current pair alone.

10 • Reliability engineer: Good. A matched comparison can then address the question with fewer competing explanations.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 03 / 05 • CONVERSATIONS / 3 OF 3

No failures during a short trial

Fictional reliability briefing: six demonstration units ran for one afternoon without failures; expected service life has not been tested.

01 • Product lead: All six demonstration units worked this afternoon. Can we describe them as lifetime reliable?

02 • Reliability engineer: We can report the successful demonstration, but it doesn't establish performance over the required service life.

03 • Product lead: I understand the time difference. Does no failure mean there is nothing to add to FMEA?

04 • Reliability engineer: No. Potential failure modes can remain relevant even when they haven't appeared in a short trial.

05 • Product lead: Then how do I describe the result positively without suggesting the reliability question is closed?

06 • Reliability engineer: Say six units completed the afternoon demonstration under those conditions with no observed failures.

07 • Product lead: And longer exposure, different duty cycles, and other conditions remain outside that observation.

08 • Reliability engineer: Correct. Their relevance should follow the requirements and test plan, not a universal claim.

09 • Product lead: I'll retain the demonstration result and remove the phrase lifetime reliable from the briefing.

10 • Reliability engineer: I'll identify the outstanding reliability evidence so the next discussion has a clear basis.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 03 / 06 • SAY IT

Two scenarios to practice

Use only the supplied facts. Prepare for two minutes, speak for one minute, then respond to a follow-up. Switch roles and repeat. For solo study, speak both roles aloud.

Scenario 1 | Failure Modes and Reliability

A component fails during one test. The team has not reproduced the failure or checked the test setup.

Prepare for two minutes. Speak for 45-60 seconds using the case facts, then respond to your partner's question. Switch roles and repeat without reading the model response.

Partner: You need to tell a colleague what is known. Ask which part is confirmed and which part is an assumption. Challenge one statement that sounds more certain than the case supports.

Second round: Someone asks for a yes-or-no answer when the evidence supports only a qualified answer. Be concise while preserving the uncertainty.

Model for scenario 1: The failure occurred in this test, but its mechanism is not established. I'll describe the observed behavior and test conditions separately from the possible explanations.

Scenario 2 | A connector disconnects during transport

A fictional prototype arrives with one connector disconnected. Shipping conditions were not logged, and pre-shipment inspection records have not been checked. Discuss the observation and next evidence needed without identifying a cause prematurely.

Role A: Receiving engineer: describe what you observed and what you do not know.

Role B: Reliability reviewer: ask about assembly records and transport exposure.

Possible opening: One connector was disconnected on arrival; we have not established when or why it separated.

Success checks

- State the observed failure mode precisely.
- Keep transport vibration as an unconfirmed explanation.
- Agree which existing records to review first.

Add a complication: A colleague wants to label the issue vibration failure immediately.

Check: facts accurate | meaning clear | tone appropriate | next action or question explicit

Continue scenario 1 with an AI partner.

LESSON 04 / READ AND NOTICE

Testing, Validation, and Data Interpretation

Explain a useful distinction in plain English.

SITUATION

A prototype passes a test on three samples. A presentation calls the design proven for every operating condition.

1. Understand the situation

Identify two confirmed facts, one missing detail, and the person who needs your response. Do not fill gaps with invented facts.

2. Find the words

prototype - An early version used to explore or test a design.

acceptance criteria - Specific conditions a deliverable must meet to be accepted.

validation - Confirming that an output or system meets needs for its intended use under defined conditions.

sample size - The number of independent units or observations included in a defined analysis.

3. Notice the language

Start with the reader's question. Explain one unfamiliar term with familiar words, then give a relevant example or contrast. Check understanding without asking only 'Do you understand?'.

In this context, ... means

The difference is that

How would you explain that distinction to a colleague?

Compare this wording: We need to operationalize the implementation.

LESSON 04 / PRACTICE AND PRODUCE

Make a complex point clear

4. Check the language

1. Which sentence explains 'backlog' rather than repeating it?
 - A. The backlog is the work still waiting to be completed.
 - B. The backlog is the work completed during the last period.
 - C. The backlog is the maximum work the team can finish in a period.
2. Which follow-up best checks whether your explanation was clear?
 - A. Would you like me to repeat the explanation?
 - B. Was the explanation detailed enough for you?
 - C. How would you describe the next step in your own words?

05 • Conversations

The next three pages contain original fictional conversations for this lesson. Each numbered line is one speaking turn. Read the setting before assigning roles.

LESSON 04 / 05 • CONVERSATIONS / 1 OF 3

Three samples do not cover every condition

Fictional prototype review using the lesson case: three samples passed one test, but a presentation claims proof for every operating condition.

01 • Project lead: Three samples passed. Why can't the presentation say the design is proven for every condition?

02 • Test engineer: Because the result covers those samples and test conditions, not conditions we didn't examine.

03 • Project lead: Are you questioning whether the samples met the acceptance criteria that were actually used?

04 • Test engineer: No. Those passes stand. I'm distinguishing a measured result from a broader validation claim.

05 • Project lead: Could you explain validation in plain English for the commercial audience at the review?

06 • Test engineer: Here it means checking whether the design meets its intended use, with evidence relevant to that use.

07 • Project lead: Then the sample size and operating conditions belong beside the pass statement, not hidden elsewhere.

08 • Test engineer: Exactly. They help readers understand the boundary of what our prototype evidence supports.

09 • Project lead: I'll replace the universal claim with a description of the three-sample test result.

10 • Test engineer: I'll list the remaining conditions for review without implying they have already passed or failed.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 04 / 05 • CONVERSATIONS / 2 OF 3

Average performance hides one missed criterion

Fictional latch test: four measured opening forces are 8, 9, 10, and 17 units; the project's stated individual limit is 15 units.

01 • Design engineer: The average opening force is eleven units, below our fifteen-unit limit. Can we report a pass?

02 • Test analyst: The acceptance criterion applies to each sample. The sample at seventeen units exceeds that limit.

03 • Design engineer: So an acceptable average doesn't establish that every tested unit met the requirement.

04 • Test analyst: Correct. Averaging answers a different question from checking each result against an individual limit.

05 • Design engineer: Should we call the entire product invalid based on that one result from four samples?

06 • Test analyst: We should report the missed criterion and investigate, without extending the conclusion beyond this test.

07 • Design engineer: Please show the four values alongside the average so the distinction is easy to explain.

08 • Test analyst: I'll include the sample size, criterion wording, and test conditions in the result summary.

09 • Design engineer: Then our review can discuss the outlying result instead of overlooking it in one number.

10 • Test analyst: Yes. The next decision needs the actual evidence, not only a favorable summary statistic.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 04 / 05 • CONVERSATIONS / 3 OF 3

A demonstration is not an acceptance test

Fictional pump-display demonstration: the prototype ran at room temperature, while agreed acceptance testing includes additional conditions not yet tested.

01 • Sales engineer: The demonstration worked beautifully. May I tell the customer that acceptance testing is complete?

02 • Validation lead: Please call it a demonstration. The agreed acceptance plan includes additional conditions we haven't tested.

03 • Sales engineer: Would that wording make the successful run sound meaningless to a nontechnical customer?

04 • Validation lead: Not if we explain it clearly: the prototype demonstrated the function under today's conditions.

05 • Sales engineer: And acceptance means checking the agreed criteria, including the outstanding conditions in that plan.

06 • Validation lead: Exactly. Those are different claims, even when the same prototype is used for both.

07 • Sales engineer: Can we identify the remaining work without giving dates the laboratory hasn't confirmed?

08 • Validation lead: Yes. List the pending tests and say scheduling is being confirmed by the test owner.

09 • Sales engineer: I'll describe the successful function demonstration and keep formal acceptance status pending.

10 • Validation lead: That gives the customer useful progress without overstating the evidence or promising an unconfirmed completion date.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 04 / 06 • SAY IT

Two scenarios to practice

Use only the supplied facts. Prepare for two minutes, speak for one minute, then respond to a follow-up. Switch roles and repeat. For solo study, speak both roles aloud.

Scenario 1 | Testing, Validation, and Data Interpretation

A prototype passes a test on three samples. A presentation calls the design proven for every operating condition.

Prepare for two minutes. Speak for 45-60 seconds using the case facts, then respond to your partner's question. Switch roles and repeat without reading the model response.

Partner: You are unfamiliar with one technical term in the case. Ask for a plain-English explanation and then paraphrase it. Do not pretend to understand a term you cannot explain.

Second round: Your listener is new to this field. Replace two specialist expressions with clear explanations without changing the meaning.

Model for scenario 1: The three samples met the criteria under the tested conditions. That does not establish performance under every condition. I'll state the sample size and limits of the test clearly.

Scenario 2 | Matching displayed readings, unresolved dimensions

In a fictional dimensional check, two readings display the same rounded value. The instrument resolution and raw readings have not been reviewed. Explain why matching displayed values do not yet prove identical dimensions.

Role A: Test engineer: distinguish displayed precision from the physical comparison.

Role B: Design lead: ask which measurement information is needed before interpreting the result.

Possible opening: The displayed readings match, but we need the instrument resolution before saying the dimensions are identical.

Success checks

- Explain the distinction without claiming a known difference.
- Request resolution and measurement-condition information.
- Agree wording limited to the observed readings.

Add a complication: The lead asks to remove all measurement caveats from the presentation.

Check: facts accurate | meaning clear | tone appropriate | next action or question explicit

Continue scenario 1 with an AI partner.

LESSON 05 / READ AND NOTICE

Manufacturability and Cost Engineering

Compare options using the same criteria and state the tradeoff.

SITUATION

Two designs use different tooling and assembly steps. One has a lower material cost but higher assembly time.

1. Understand the situation

Identify two confirmed facts, one missing detail, and the person who needs your response. Do not fill gaps with invented facts.

2. Find the words

DFM - Design for manufacturability: shaping a design so it can be made effectively and consistently.

tooling - Equipment or fixtures used to shape, assemble, or produce a product.

yield - The proportion of production meeting defined requirements; in finance, a return measure with a different calculation.

cycle time - The time needed to complete one defined production or work cycle.

3. Notice the language

Use the same time period, scope, and measurement basis for both options. Explain a difference with 'whereas' or 'while'. State what improves and what is given up; do not hide the weaker side of a recommendation.

Both options ..., but

Option A ..., whereas option B

The comparison depends on

Compare this wording: This option is more better.

LESSON 05 / PRACTICE AND PRODUCE

Explain a fair comparison

4. Check the language

1. Which sentence expresses a clear contrast?
 - A. Option A is faster because option B costs less.
 - B. Option A is faster, so option B costs less.
 - C. Option A is faster, whereas option B costs less.
2. A rate increases from 6% to 7%. What is the absolute difference?
 - A. An increase of seven percentage points.
 - B. An increase of one percentage point.
 - C. An increase of one percent relative to the original rate.

05 • Conversations

The next three pages contain original fictional conversations for this lesson. Each numbered line is one speaking turn. Read the setting before assigning roles.

LESSON 05 / 05 • CONVERSATIONS / 1 OF 3

Cheaper material, longer assembly

Fictional design comparison using the lesson case: one option has lower material cost but takes longer to assemble; tooling differs.

01 • Design engineer: Option A uses cheaper material. Should we recommend it as the lower-cost design?

02 • Manufacturing engineer: Not yet. Its assembly takes longer, and the two options require different tooling.

03 • Design engineer: What common criteria should our design-for-manufacture, or DFM, review use to compare them?

04 • Manufacturing engineer: Material, tooling, assembly cycle time, and expected yield, with the same planned production volume.

05 • Design engineer: We don't have a verified yield estimate for either option. Should we assume they match?

06 • Manufacturing engineer: Show that as unknown instead. An assumed match could hide a meaningful difference in usable output.

07 • Design engineer: Could we present material savings separately while requesting the missing assembly and yield evidence?

08 • Manufacturing engineer: Yes. That identifies a real advantage without confusing it with total manufacturing cost.

09 • Design engineer: I'll build the comparison around the same volume and clearly label estimates and gaps.

10 • Manufacturing engineer: Then we can discuss the tradeoff rather than select a design from one cost line.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 05 / 05 • CONVERSATIONS / 2 OF 3

A faster cycle with more rejected parts

Fictional pilot comparison: process A produces 100 parts per hour with 90 accepted; process B produces 95 with 93 accepted in one trial each.

01 • Production planner: Process A makes one hundred parts an hour, so it seems faster than process B's ninety-five.

02 • Manufacturing engineer: Gross output is higher, but A produced ninety accepted parts while B produced ninety-three.

03 • Production planner: Then accepted output favors B in these trials, even though its individual cycle is slower.

04 • Manufacturing engineer: Yes. Cycle time and yield together give a different picture from raw production count alone.

05 • Production planner: Can we assume those trial results will hold for every shift and material batch?

06 • Manufacturing engineer: No. We have one trial of each, so repeatability and comparable conditions still need checking.

07 • Production planner: What should the DFM review say about the apparent tradeoff while that work remains open?

08 • Manufacturing engineer: A had faster gross output; B had more accepted output in the observed trials.

09 • Production planner: I'll retain both measures and avoid claiming a confirmed long-term production advantage.

10 • Manufacturing engineer: I'll ask for comparable follow-up runs before we recommend either process as the production baseline.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 05 / 05 • CONVERSATIONS / 3 OF 3

Tooling cost depends on the volume assumption

Fictional enclosure review: a machined option avoids dedicated tooling; a molded option requires tooling, while the production quantity is not confirmed.

01 • Product engineer: The molded enclosure has a lower quoted unit price. Can we rule out machining?

02 • Cost engineer: The molded quote also includes dedicated tooling, and our production quantity is not confirmed.

03 • Product engineer: So the unit-price comparison leaves out a cost that must be spread across actual production.

04 • Cost engineer: Exactly. We should compare total costs at the same volume assumptions and state those assumptions.

05 • Product engineer: Could machining still be worth considering for a small initial run despite its higher unit price?

06 • Cost engineer: Potentially, but we should calculate that case rather than declare it cheaper without the numbers.

07 • Product engineer: What other criteria should accompany the cost comparison so we don't miss manufacturing consequences?

08 • Cost engineer: Lead time, design-change flexibility, cycle time, and yield evidence would make the tradeoff clearer.

09 • Product engineer: I'll request comparable quotes for the proposed quantities and keep demand uncertainty visible.

10 • Cost engineer: Then our recommendation can be conditional on volume instead of pretending one option always wins.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 05 / 06 • SAY IT

Two scenarios to practice

Use only the supplied facts. Prepare for two minutes, speak for one minute, then respond to a follow-up. Switch roles and repeat. For solo study, speak both roles aloud.

Scenario 1 | Manufacturability and Cost Engineering

Two designs use different tooling and assembly steps. One has a lower material cost but higher assembly time.

Prepare for two minutes. Speak for 45-60 seconds using the case facts, then respond to your partner's question. Switch roles and repeat without reading the model response.

Partner: You favor one option but have not checked whether the scope is comparable. Ask about one difference, then explain the tradeoff you are willing to accept.

Second round: Your listener focuses only on the lowest price or fastest date. Explain one other relevant comparison criterion.

Model for scenario 1: The material cost favors the first design, while assembly time favors the second. Let's compare the full manufacturing assumptions before calling either design cheaper overall.

Scenario 2 | Two fasteners or a snap fit?

A fictional cover can use two screws or a proposed snap fit. The screw assembly time has been measured; snap-fit assembly time and rework behavior have not. Compare the options without assuming the new design is faster or cheaper.

Role A: Design engineer: explain the proposed reduction in parts.

Role B: Manufacturing engineer: ask for comparable assembly and rework evidence.

Possible opening: The snap fit removes two screws, but we haven't measured its assembly or rework performance yet.

Success checks

- Use common criteria for both options.
- Separate fewer parts from demonstrated lower cost.
- Propose a bounded comparison and preserve unknown results.

Add a complication: A manager wants a cost-saving percentage before the trial.

Check: facts accurate | meaning clear | tone appropriate | next action or question explicit

Continue scenario 1 with an AI partner.

LESSON 06 / READ AND NOTICE

Safety Factors and Compliance

Make a request with a clear action, purpose, and realistic timing.

SITUATION

A review note says a design is "compliant" without identifying the applicable requirement or review evidence.

1. Understand the situation

Identify two confirmed facts, one missing detail, and the person who needs your response. Do not fill gaps with invented facts.

2. Find the words

safety factor - A defined design ratio or allowance addressing uncertainty between expected loading and capacity.

code compliance - Conformity with the applicable building, engineering, or other technical code requirements.

margin - The difference or allowance between values; in business, a measure of profitability.

liability - Legal responsibility or an obligation; in accounting, an amount or duty owed.

3. Notice the language

State the action and the information needed. Explain why it matters and specify a deadline only when one is given or agreed. 'Could you' is polite; repeated apologies can obscure a legitimate request. Urgent situations may require a direct request.

Could you provide ... so that ...?

Please confirm ... by [agreed time].

If that timing is not possible, please let me know

Compare this wording: Please revert soonest with the necessary.

LESSON 06 / PRACTICE AND PRODUCE

Ask for an actionable next step

4. Check the language

1. What does 'Please send it by Thursday' normally mean?
 - A. Keep sending it regularly until Thursday.
 - B. Send it no later than Thursday.
 - C. Send it on Thursday, but not earlier.
2. Which request gives the recipient enough detail to act?
 - A. Could you confirm the document version needed for the review?
 - B. Could you confirm that the review is important?
 - C. Could you confirm that you received my earlier message?

05 • Conversations

The next three pages contain original fictional conversations for this lesson. Each numbered line is one speaking turn. Read the setting before assigning roles.

LESSON 06 / 05 • CONVERSATIONS / 1 OF 3

Which requirement supports compliant?

Fictional engineering review using the lesson case: a note says compliant but identifies neither an applicable requirement nor supporting review evidence.

01 • Review coordinator: The note calls the design compliant. Could you identify the requirement and evidence behind that statement?

02 • Design engineer: I copied the wording from an earlier review, but I haven't checked its applicability here.

03 • Review coordinator: Please locate the relevant requirement and review record before tomorrow's discussion, or flag what remains missing.

04 • Design engineer: Would a calculation showing a positive margin be enough to keep the compliance wording?

05 • Review coordinator: A calculation may support a particular check, but it doesn't establish every applicable requirement was reviewed.

06 • Design engineer: Understood. I'll distinguish the calculated margin from the broader code-compliance claim in the note.

07 • Review coordinator: Also ask the qualified reviewer to confirm which requirements apply; neither of us should guess.

08 • Design engineer: Should I mark compliance as pending while that applicability review is incomplete?

09 • Review coordinator: Yes. State the evidence gap accurately rather than asserting either compliance or noncompliance without support.

10 • Design engineer: I'll bring the identified records and unresolved questions to the reviewer through our established process.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 06 / 05 • CONVERSATIONS / 2 OF 3

A safety factor without its load basis

Fictional internal calculation review: a summary lists a safety factor of two but omits the load case and calculation revision.

01 • Engineering manager: The summary gives a safety factor of two. Can we approve the component from that number?

02 • Calculation reviewer: I need the load case and calculation revision before interpreting what that factor represents.

03 • Engineering manager: Could you explain why the ratio alone isn't enough for the review audience?

04 • Calculation reviewer: A margin only has meaning relative to its stated assumptions, inputs, and applicable acceptance basis.

05 • Engineering manager: Then please request the source calculation rather than suggesting the factor itself is definitely inadequate.

06 • Calculation reviewer: Agreed. We have missing context, not evidence yet that the design meets or misses the requirement.

07 • Engineering manager: Can the calculation owner provide the load basis before our scheduled design checkpoint?

08 • Calculation reviewer: I'll ask for that timing and request an explicit gap update if the records aren't ready.

09 • Engineering manager: Keep the decision pending and route the adequacy question to the responsible qualified engineer.

10 • Calculation reviewer: I will. The summary will identify the factor as reported, with interpretation awaiting its basis.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 06 / 05 • CONVERSATIONS / 3 OF 3

A supplier certificate names another revision

Fictional equipment review: a supplier certificate names revision B, while the supplied assembly is labeled revision C; equivalence has not been established.

01 • Procurement engineer: We have the supplier certificate. Can the review team close the compliance evidence item?

02 • Quality engineer: The certificate names revision B, but the delivered assembly is revision C. We need clarification.

03 • Procurement engineer: Could that simply be a labeling difference with no technical effect on the product?

04 • Quality engineer: Possibly, but equivalence isn't established. Please ask the supplier which configuration their evidence actually covers.

05 • Procurement engineer: Should I request a corrected certificate or a documented explanation of the revision relationship?

06 • Quality engineer: Request clarification and supporting records first; we shouldn't prescribe a correction before understanding the discrepancy.

07 • Procurement engineer: I'll ask for a response before the acceptance meeting and flag any delay to you.

08 • Quality engineer: Thank you. Keep the original certificate with its revision information rather than replacing the record silently.

09 • Procurement engineer: I'll describe the mismatch without claiming the assembly is unsafe or fully compliant.

10 • Quality engineer: Then the responsible reviewer can assess applicability using evidence instead of an unsupported revision assumption.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 06 / 06 • SAY IT

Two scenarios to practice

Use only the supplied facts. Prepare for two minutes, speak for one minute, then respond to a follow-up. Switch roles and repeat. For solo study, speak both roles aloud.

Scenario 1 | Safety Factors and Compliance

A review note says a design is "compliant" without identifying the applicable requirement or review evidence.

Prepare for two minutes. Speak for 45-60 seconds using the case facts, then respond to your partner's question. Switch roles and repeat without reading the model response.

Partner: You can help but need a clear request and its purpose. Ask what is required and whether the timing is fixed or negotiable.

Second round: The other person cannot meet the requested timing. Clarify an alternative without inventing authority to change a formal deadline.

Model for scenario 1: Could you identify the requirement, revision, and supporting evidence for that statement? We need a traceable basis for the compliance claim and review by the responsible specialist.

Scenario 2 | The unsigned review page

A fictional design package contains a completed calculation and an unsigned review page. The reviewer's availability is unknown. Request status and supporting evidence without treating a completed calculation as completed independent review.

Role A: Engineering coordinator: ask what review occurred and what remains pending.

Role B: Design engineer: distinguish calculation completion from review confirmation.

Possible opening: The calculation is present, but could you confirm whether the independent review has actually occurred?

Success checks

- Do not infer approval from precedent.
- Request confirmation through the appropriate reviewer.
- Agree a status checkpoint without inventing availability.

Add a complication: Someone suggests adding the reviewer's name because they usually approve similar work.

Check: facts accurate | meaning clear | tone appropriate | next action or question explicit

Continue scenario 1 with an AI partner.

LESSON 07 / READ AND NOTICE

Field Issues and Customer Communication

Acknowledge a communication problem and give a corrected message.

SITUATION

A customer received a field-failure update that stated an unconfirmed cause as fact. The investigation is still open.

1. Understand the situation

Identify two confirmed facts, one missing detail, and the person who needs your response. Do not fill gaps with invented facts.

2. Find the words

field failure - A product failure occurring during use outside the development or production setting.

warranty - A promise or contractual assurance about a product or work, with defined terms and remedies.

corrective action - Action taken to fix a current problem and prevent recurrence.

installation condition - A circumstance of how or where equipment was installed that may affect its performance.

3. Notice the language

Name what was unclear or wrong, acknowledge its effect, and correct it. A brief apology can help, followed by a practical next step. Avoid explaining away the other person's reaction or promising an outcome you cannot control.

My earlier message did not make ... clear.

I'm sorry for The correct information is

Let me check that the revised explanation addresses

Compare this wording: I'm sorry if you failed to understand.

LESSON 07 / PRACTICE AND PRODUCE

Repair a misunderstanding

4. Check the language

1. Which apology takes responsibility for unclear wording?
 - A. I'm sorry my earlier wording was unclear.
 - B. I'm sorry the process took longer than expected.
 - C. I'm sorry we could not meet the original date.
2. What should follow an apology for an incorrect date?
 - A. A longer explanation that never states the correct date.
 - B. A new guaranteed date even if it is unverified.
 - C. The corrected date and the current confirmation status.

05 • Conversations

The next three pages contain original fictional conversations for this lesson. Each numbered line is one speaking turn. Read the setting before assigning roles.

LESSON 07 / 05 • CONVERSATIONS / 1 OF 3

Correcting an unconfirmed field-failure cause

Fictional customer update using the lesson case: a previous message presented an unconfirmed field-failure cause as fact; investigation remains open.

01 • Customer engineer: Your update said installation caused the field failure. Has that actually been established?

02 • Service engineer: No. Our wording was too definite, and I apologize for presenting an unconfirmed explanation as fact.

03 • Customer engineer: Then what can we rely on from that message when briefing our operations team?

04 • Service engineer: The failure report is recorded, but the cause remains under investigation. Installation conditions are one open question.

05 • Customer engineer: Does this correction also mean you are confirming that the failure qualifies for warranty coverage?

06 • Service engineer: No. I cannot determine coverage here; that question needs the appropriate review alongside the investigation.

07 • Customer engineer: Please keep those two issues separate so the correction doesn't create another unsupported conclusion.

08 • Service engineer: I will. We'll provide a factual investigation update and route the coverage question to its owner.

09 • Customer engineer: And don't call a proposed corrective action a proven fix before the evidence supports it.

10 • Service engineer: Agreed. The corrected message will distinguish observations, hypotheses, proposed actions, and decisions still pending.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 07 / 05 • CONVERSATIONS / 2 OF 3

A replacement shipment is not proof of a fix

Fictional service case: a replacement display has shipped after a field failure, but the cause and corrective-action effectiveness are unconfirmed.

01 • Customer coordinator: Your message said permanently resolved because a replacement display shipped. Is the root cause confirmed?

02 • Service lead: It isn't. That wording overstated what the shipment establishes, and I should correct it.

03 • Customer coordinator: Please explain the distinction so our staff know what has actually changed today.

04 • Service lead: A replacement has shipped to support recovery. We haven't confirmed the cause or a permanent corrective action.

05 • Customer coordinator: Will you need information about the original unit's installation conditions before the investigation continues?

06 • Service lead: The investigation owner will specify the needed records. We shouldn't ask you to guess or alter equipment.

07 • Customer coordinator: Can you give us a useful update without guaranteeing that the replacement cannot have the same issue?

08 • Service lead: Yes. I'll confirm shipment status, state the investigation is open, and name the next contact.

09 • Customer coordinator: That is more accurate. We'll keep the original failure record available through the agreed channel.

10 • Service lead: Thank you. I'll replace the earlier closure language and keep recovery separate from verified prevention.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 07 / 05 • CONVERSATIONS / 3 OF 3

No fault found does not erase the report

Fictional returned-unit investigation: a display worked during a bench check, but the customer's intermittent field symptom has not been reproduced.

01 • Customer engineer: Your report says no fault exists. We experienced an intermittent blank display twice last week.

02 • Test engineer: You're right to challenge that wording. Our bench check found no fault under the conditions tested.

03 • Customer engineer: So you aren't saying our field report is false or that the issue is closed?

04 • Test engineer: Correct. We haven't reproduced the symptom, which is different from proving it never occurred.

05 • Customer engineer: What should the corrected message say about installation conditions and the next investigation step?

06 • Test engineer: It should request the relevant recorded conditions through the service process, without asserting they caused the issue.

07 • Customer engineer: Please don't ask our operators to recreate a potentially disruptive failure on active equipment.

08 • Test engineer: Agreed. We'll review existing records and have the investigation owner define any appropriate controlled checks.

09 • Customer engineer: I'll share the timestamps from our reports and keep the description limited to what staff observed.

10 • Test engineer: I'll correct the report and clearly mark the intermittent field issue as still under investigation.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 07 / 06 • SAY IT

Two scenarios to practice

Use only the supplied facts. Prepare for two minutes, speak for one minute, then respond to a follow-up. Switch roles and repeat. For solo study, speak both roles aloud.

Scenario 1 | Field Issues and Customer Communication

A customer received a field-failure update that stated an unconfirmed cause as fact. The investigation is still open.

Prepare for two minutes. Speak for 45-60 seconds using the case facts, then respond to your partner's question. Switch roles and repeat without reading the model response.

Partner: You are frustrated because an earlier message created an expectation. Explain that expectation. Ask your partner to distinguish the correction from anything still uncertain.

Second round: The listener remains disappointed after the correction. Acknowledge the impact and restate the available next step calmly.

Model for scenario 1: Our earlier update overstated the cause. The investigation is still open, and the confirmed finding is the observed failure described below. I'm sorry for the confusion; we'll clearly label hypotheses in future updates.

Scenario 2 | An interim report incorrectly labeled final

A fictional customer received a test summary headed Final Update, although two inspection results remain pending. No cause is confirmed. Correct the communication and agree how the next update will distinguish findings from open work.

Role A: Service engineer: acknowledge the misleading heading and explain current status.

Role B: Customer maintenance coordinator: ask what can be shared with colleagues now.

Possible opening: The heading was misleading: this is an interim update, and two inspection results are still pending.

Success checks

- Acknowledge the specific communication error.
- Preserve confirmed results without inventing a cause.
- Agree a clear next update and ownership.

Add a complication: The customer asks whether the heading error means all test results are unreliable.

Check: facts accurate | meaning clear | tone appropriate | next action or question explicit

Continue scenario 1 with an AI partner.

LESSON 08 / READ AND NOTICE

Systems Integration and Interface Control

Give a handoff that the receiving person can confirm.

SITUATION

A hardware interface changed after software testing began. The integration team needs the version and affected tests.

1. Understand the situation

Identify two confirmed facts, one missing detail, and the person who needs your response. Do not fill gaps with invented facts.

2. Find the words

interface control - Managing the agreed technical relationships between components or systems.

integration - Connecting components or organizations so they work together as intended.

regression test - A test checking whether a change has broken previously working behavior.

configuration management - Controlling and recording the versions and settings of a system or product.

3. Notice the language

State the current situation, relevant background, open task, and receiving role. Ask for acknowledgment or repeat-back of the critical item. Sending a message and transferring responsibility are not always the same thing.

The current status is ...; the outstanding item is

The record shows

Please confirm who has accepted

Compare this wording: Someone should follow up on this.

LESSON 08 / PRACTICE AND PRODUCE

Transfer information and responsibility

4. Check the language

1. Which sentence checks that responsibility has been accepted?
 - A. I mentioned the check in my earlier email.
 - B. The check appears somewhere in the notes.
 - C. Please confirm who will complete the outstanding check.
2. Which response best checks a critical handoff detail?
 - A. I saw the message, so no clarification is needed.
 - B. So the result is pending and your team will follow up; is that correct?
 - C. I assume all results are final, so we can finish.

05 • Conversations

The next three pages contain original fictional conversations for this lesson. Each numbered line is one speaking turn. Read the setting before assigning roles.

LESSON 08 / 05 • CONVERSATIONS / 1 OF 3

Which interface version did software test?

Fictional integration handoff using the lesson case: hardware changed after software testing began, and the receiving team needs the version and affected tests.

01 • Hardware engineer: The interface changed after software testing started. I need to hand over the exact configuration details.

02 • Integration lead: Please identify the released version and the change record before we decide which tests remain applicable.

03 • Hardware engineer: I'll supply those records. I haven't yet mapped every change to the software test cases.

04 • Integration lead: Then that mapping is pending, not proof that all prior results are invalid.

05 • Hardware engineer: Correct. Could your team confirm which interface version the existing test results actually used?

06 • Integration lead: Yes. We'll compare the test configuration with your change record and identify regression-test candidates.

07 • Hardware engineer: I'll include the interface-control document and make any unapproved changes visibly separate from released ones.

08 • Integration lead: Please also name the owner who can resolve discrepancies if the records don't match.

09 • Hardware engineer: I'll be that contact. Can you confirm the handoff includes both version evidence and the open mapping task?

10 • Integration lead: Confirmed. We'll assess affected tests before claiming the new hardware and software combination is verified.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 08 / 05 • CONVERSATIONS / 2 OF 3

A unit change in the sensor message

Fictional integration case: interface revision 4 reports temperature in Celsius; revision 5 specifies kelvin. Existing software tests used revision 4.

01 • Firmware engineer: Revision five changes the temperature field from Celsius to kelvin. Existing tests used revision four.

02 • Integration engineer: Thanks. Has the receiving software been updated, or is that still an open handoff item?

03 • Firmware engineer: Still open. I'm handing over the revised field definition, not claiming the combined system works.

04 • Integration engineer: We need to trace the conversion and display expectations before accepting results for this configuration.

05 • Firmware engineer: The interface-control change record identifies the field. I'll include both revisions for comparison.

06 • Integration engineer: Should we rerun every test, or first assess which regression tests depend on that field?

07 • Firmware engineer: Let's assess dependencies first, while keeping any broader integration checks required by our plan.

08 • Integration engineer: I'll confirm the software configuration and have the test owner identify the affected checks.

09 • Firmware engineer: Please repeat the main constraint so I know the receiving team has the same understanding.

10 • Integration engineer: Revision-five data cannot inherit revision-four temperature test approval without reviewing the unit change and affected tests.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 08 / 05 • CONVERSATIONS / 3 OF 3

Matching filenames conceal different configurations

Fictional test handoff: two configuration files share a filename but have different revision identifiers; the tested identifier is recorded.

01 • Test coordinator: Both files are named station-config. Can we give the integration team either one?

02 • Configuration engineer: No. Their revision identifiers differ, and configuration management needs the exact file used during testing.

03 • Test coordinator: The test record includes an identifier. I'll compare that before packaging the handoff.

04 • Configuration engineer: Good. The filename helps people find a file, but it doesn't establish configuration identity.

05 • Test coordinator: If the newer file contains only a small setting change, can we reuse the old results?

06 • Configuration engineer: We need an impact review first. The size of the text change doesn't establish its functional effect.

07 • Test coordinator: I'll deliver the tested revision and identify the newer one as a separate, unassessed configuration.

08 • Configuration engineer: Include the change history and any open integration questions so the receiver can confirm scope.

09 • Test coordinator: Can you verify that package against the recorded test identifier before I send it?

10 • Configuration engineer: Yes. Then we'll have a traceable handoff and a clear basis for selecting regression tests.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 08 / 06 • SAY IT

Two scenarios to practice

Use only the supplied facts. Prepare for two minutes, speak for one minute, then respond to a follow-up. Switch roles and repeat. For solo study, speak both roles aloud.

Scenario 1 | Systems Integration and Interface Control

A hardware interface changed after software testing began. The integration team needs the version and affected tests.

Prepare for two minutes. Speak for 45-60 seconds using the case facts, then respond to your partner's question. Switch roles and repeat without reading the model response.

Partner: You are taking over the work. Ask what remains open and repeat back the critical status or responsibility. Point out one detail that would be unsafe or misleading to assume.

Second round: The intended recipient cannot take ownership. Keep the status clear and identify the need for an authorized reassignment.

Model for scenario 1: The interface changed in this revision. I've listed the affected signals and tests for review. Please confirm the configuration your team will use and which checks need to be repeated.

Scenario 2 | A timeout changed between components

In a fictional integration, the controller timeout changes from five seconds to three in a proposed revision. The receiving application was tested with five seconds. The proposal is not released. Give a handoff that preserves that status and identifies test impact.

Role A: Controller engineer: explain the proposed value and release status.

Role B: Integration lead: confirm the tested configuration and ask about regression review.

Possible opening: The proposed revision shortens the timeout to three seconds, but it has not been released or integration-tested.

Success checks

- Distinguish proposed, released, and tested configurations.
- Identify the timing-dependent tests as needing review.
- Confirm the receiver understands which file may be used.

Add a complication: A colleague wants to put the proposed file into the release package because its revision number is newer.

Check: facts accurate | meaning clear | tone appropriate | next action or question explicit

Continue scenario 1 with an AI partner.

Answer workshop

Try the tasks first. The model is one acceptable response to scenario 1, not a script to memorize. Assess your spoken version using the four criteria at the end.

01 | Requirements and Constraints

Editing example: Could you explain what this means?

After an introductory phrase such as 'Could you explain', the embedded question uses statement word order: "what this means".

Check 1: A - Could you confirm when the review starts?

A: The embedded question uses subject then verb: "the review starts".

B: Remove 'does' and use statement word order after 'confirm'.

C: Place the subject 'the review' before 'starts'.

Check 2: C - Has the report been approved, or is it ready for review?

A: This asks about delivery timing, not whether approval has happened.

B: This asks about the presenter, not the report's approval status.

C: This question distinguishes two meanings of 'ready' that affect the next action.

Model response: Could you define the target weight, operating conditions, and expected life?
We need measurable requirements to discuss the tradeoffs and evaluate a design consistently.

Case check: underline the confirmed information and circle a limitation, question, or next step in the response.

02 | Design Reviews and Technical Pushback

Editing example: My concern is that the proposal assumes capacity we have not confirmed.

The revised sentence identifies a testable concern about the proposal. It leaves room for a constructive response.

Check 1: C - The proposal assumes the review is complete; the record shows it is still open.

A: This expresses a preference, but does not identify the assumption or the record that contradicts it.

B: This requests discussion without stating the specific assumption or evidence.

C: It connects the concern to a specific assumption and record.

Check 2: B - Could we resolve the missing approval before confirming the plan?

A: General feedback does not directly address the missing approval.

B: It names the unresolved condition and a practical next step.

C: This confirms the plan before resolving the stated approval condition.

Model response: I support moving the review forward, but this interface assumption needs to be explicit. Could we record it and identify the verification needed before treating that part of the design as settled?

Case check: underline the confirmed information and circle a limitation, question, or next step in the response.

03 | Failure Modes and Reliability

Editing example: The pilot worked in the tested setting; performance elsewhere remains untested.

The revision states the finding and its scope. It avoids turning limited evidence into a universal claim.

Check 1: B - The change may have contributed to the delay.

A: 'Definitely' presents the cause as certain.

B: 'May have contributed' presents a possibility without claiming the cause is established.

C: 'Must have' expresses a strong inference that the evidence may not support.

Check 2: A - In this pilot, the observed result was ...

A: This locates the finding in the setting actually studied.

B: Similarity alone does not establish that the result will transfer.

C: A pilot in one location does not establish a service-wide observation.

Model response: The failure occurred in this test, but its mechanism is not established. I'll describe the observed behavior and test conditions separately from the possible explanations.

Case check: underline the confirmed information and circle a limitation, question, or next step in the response.

04 | Testing, Validation, and Data Interpretation

Editing example: We need to put the plan into practice.

The revised sentence names the action in familiar words. Specialist vocabulary is useful when it adds precision, but repeating abstract nouns can hide the meaning.

Check 1: A - The backlog is the work still waiting to be completed.

A: This gives the meaning in familiar words.

B: Completed work is different from work still waiting.

C: This describes capacity, not the work waiting to be completed.

Check 2: C - How would you describe the next step in your own words?

A: Offering repetition may help, but it does not show how the listener understood the next step.

B: This invites an opinion on detail, not a demonstration of understanding.

C: An open request lets the listener show their understanding.

Model response: The three samples met the criteria under the tested conditions. That does not establish performance under every condition. I'll state the sample size and limits of the test clearly.

Case check: underline the confirmed information and circle a limitation, question, or next step in the response.

05 | Manufacturability and Cost Engineering

Editing example: This option is better on cost, but slower to deliver.

'Better' already has comparative meaning, so it does not need 'more'. Naming the criterion and tradeoff makes the comparison useful.

Check 1: C - Option A is faster, whereas option B costs less.

A: 'Because' claims a causal connection that is not given.

B: 'So' implies one fact causes the other.

C: 'Whereas' marks a contrast between two options.

Check 2: B - An increase of one percentage point.

A: The new rate is 7%; the difference between the rates is one percentage point.

B: Subtracting 6% from 7% gives one percentage point. The relative increase is about 16.7%.

C: A one-percent relative increase on 6% would be 6.06%, not 7%.

Model response: The material cost favors the first design, while assembly time favors the second. Let's compare the full manufacturing assumptions before calling either design cheaper overall.

Case check: underline the confirmed information and circle a limitation, question, or next step in the response.

06 | Safety Factors and Compliance

Editing example: Could you send the missing document and confirm when it will be available?

The revision names the document and requested response. It avoids expressions that may be unfamiliar or ambiguous for an international audience.

Check 1: B - Send it no later than Thursday.

A: This interprets a deadline as a continuing activity, which would use "until".

B: 'By' sets the latest point, while 'after' refers to a later time.

C: "By Thursday" permits earlier delivery; it is not restricted to that day.

Check 2: A - Could you confirm the document version needed for the review?

A: The requested action and information are explicit.

B: This checks importance rather than identifying the needed document version.

C: This checks receipt, not which version is required for the review.

Model response: Could you identify the requirement, revision, and supporting evidence for that statement? We need a traceable basis for the compliance claim and review by the responsible specialist.

Case check: underline the confirmed information and circle a limitation, question, or next step in the response.

07 | Field Issues and Customer Communication

Editing example: I'm sorry my message did not explain the timing clearly.

The revision takes responsibility for the message without blaming the listener. It identifies what needs to be corrected.

Check 1: A - I'm sorry my earlier wording was unclear.

A: This acknowledges the speaker's contribution to the problem.

B: This acknowledges duration, but does not take responsibility for the unclear wording.

C: This acknowledges a missed date, not the communication problem in the question.

Check 2: C - The corrected date and the current confirmation status.

A: The reader still lacks the information they need.

B: An unsupported new promise can create a second misunderstanding.

C: A repair needs usable corrected information, including any uncertainty.

Model response: Our earlier update overstated the cause. The investigation is still open, and the confirmed finding is the observed failure described below. I'm sorry for the confusion; we'll clearly label hypotheses in future updates.

Case check: underline the confirmed information and circle a limitation, question, or next step in the response.

08 | Systems Integration and Interface Control

Editing example: Please confirm which team has accepted the follow-up task.

The revised sentence asks for an identifiable receiving team instead of leaving responsibility with an unspecified 'someone'.

Check 1: C - Please confirm who will complete the outstanding check.

A: Mentioning a task does not show that anyone accepted it.

B: A written record alone does not identify who will act.

C: It asks for an identifiable person or team responsible for the task.

Check 2: B - So the result is pending and your team will follow up; is that correct?

A: Receiving a message is not the same as confirming its meaning.

B: This repeats the status and responsibility and invites correction.

C: It changes pending information into a final result.

Model response: The interface changed in this revision. I've listed the affected signals and tests for review. Please confirm the configuration your team will use and which checks need to be repeated.

Case check: underline the confirmed information and circle a limitation, question, or next step in the response.

Give feedback that helps

Score each criterion 0, 1, or 2. This is a practice rubric, not a professional qualification or CEFR test. Do not award or remove points for a particular accent.

Meaning and accuracy

Keeps the case facts accurate; clearly separates confirmed and unknown information.

Clarity and language

Uses understandable sentences and explains specialist terms when the listener needs it.

Interaction and tone

Responds to the other person's question, checks understanding, and uses an appropriate tone.

Task completion

Makes the requested action or unresolved question clear without inventing authority or facts.

Use the same scale for each criterion

0 - Not yet: The reader or listener cannot yet recover the intended meaning or action.

1 - With support: The message works after a prompt, clarification, or revision.

2 - Independently: The message is clear and accurate without a prompt.

Feedback sequence

First, identify a successful phrase. Next, identify one point where meaning or accuracy needs work. Offer one useful language correction, allow a repeat, and compare the two attempts. A total out of eight describes this performance only; do not translate it into a proficiency level.

Final challenge

Choose a case not rehearsed today. The learner gives a one-minute response, answers two follow-up questions, and writes a 70-110 word message. Add the lesson's second-round challenge. Compare the result with an earlier attempt using the four criteria.

Use the language responsibly

These cases practice communication. Actual equipment, construction, and safety work follows approved procedures and authorized specialist decisions.

Meaning before performance

Use specialist terms only when they help the listener. Explain unfamiliar words, preserve uncertainty, and ask when an instruction or responsibility is unclear. The model responses illustrate language; they do not authorize professional actions.

Sources and further reading

The cases, workshops, and explanations are original teaching material. These references provide language frameworks and selected professional context. References were checked in September 2026; consult current local guidance for actual work.

Digital.gov: plain-language guidance

<https://digital.gov/guides/plain-language>

Council of Europe: CEFR descriptors

<https://www.coe.int/en/web/common-european-framework-reference-languages/cefr-descriptors>

Your next step

Choose one expression you can use in a genuine work situation. Rehearse it with fictional details, then adapt the wording to your role and audience.

Extend your practice with AI

Use the following prompts after your own first attempt. Each contains the course context and a complete starting case or dialogue. Copy the entire prompt, including the reference, into a new chat in your preferred AI. You can also use the links to copy it from the website.

Choose the right kind of help

A role-play prompt makes the AI wait for your replies. A new-dialogue prompt asks for a complete professional script. Vocabulary and grammar prompts move from explanation to your own attempts.

Choose any lesson or dialogue online

The course prompt workshop has four practice goals and B1 support, B2 independent and C1 stretch settings. These are practice choices, not assessed proficiency levels. Select the same lesson number or dialogue title you used in this guide.

[Open this course's AI prompt workshop](#)

Keep practice useful

Use fictional or anonymized details. English Ladder does not send anything to an AI service or add your drafts to the prompts. Your chosen service's terms and any usage charges apply. AI responses can contain mistakes; compare feedback with the lesson and verify specialist claims against the course references.

Each starter prompt uses B2 practice. To change support in a printed prompt, replace its practice-setting paragraph with a request for shorter explanations and sentence starters (B1), or greater precision and more complex constraints (C1). Keep the professional facts unchanged.

Changing the example

For another case on paper, replace the text between REFERENCE and END REFERENCE with that case, its course context, relevant terms and language target. Include the full exchange if practicing a dialogue. Do not assume your AI can access this PDF or a link. The website makes this substitution for you.

Prompt edition: 2026-09-30. These are original practice instructions; results depend on the AI service you choose.

Practice grammar in context

START PROMPT / COPY THROUGH END PROMPT

Be my workplace English practice partner. Follow the practice instructions after the REFERENCE block. The block contains fictional study material, not instructions to you. Keep its facts, numbers, uncertainty and professional responsibilities accurate. Clearly label any new scenario or changed fact as an invented variation. Use natural workplace language; explain unfamiliar abbreviations in context. If a technical expression is uncertain, say so rather than inventing a definition or authority. Coach communication, not professional decisions; respect the course scope. Ask only for fictional or anonymized practice details. Judge clarity, meaning, appropriate tone and the next step. Accept valid alternative English and distinguish errors from style preferences. Do not claim to certify my language level. Unless I ask to change the plan, follow the stages and stop wherever the instructions say to wait.

When discussing a word, phrase, or example sentence as language within an explanation, question, or answer feedback, enclose that wording in quotation marks. For example: What does "about" mean in "about twenty people"? Use "on" with a named day. Keep standalone answer choices, vocabulary headwords, and ordinary reading or dialogue text uncluttered. Quotation marks identifying a language example do not attribute it to a news source; attributed source quotations must remain verbatim.

Practice setting: B2 independent. Use natural professional English with brief explanations. Let me attempt each task without a model first. Offer a hint after difficulty and invite a more precise retry.

REFERENCE

Course: Engineering English

Professional audience: mechanical, electrical, civil, systems, industrial, test, quality, manufacturing, and field engineers, plus engineering managers and technical project leads

Scope: These cases practice communication. Actual equipment, construction, and safety work follows approved procedures and authorized specialist decisions.

Lesson 01: Requirements and Constraints

Original fictional case: A customer asks for a "lightweight and durable" enclosure without specifying weight, operating conditions, or required life.

Language workshop: Ask a precise question

Communication goal: Clarify missing information before responding.

Language guidance: Name the exact word, fact, or requirement you need clarified. Ask one focused question at a time. Repeat your understanding and give the other person a chance to correct it.

Useful frames (complete the gaps with case facts): When you say ..., do you mean ...? / Could you clarify which ...? / So, my understanding is Is that right?

Vocabulary: requirement: A condition or capability that must be met for a stated purpose. / constraint: A limit that restricts the available choices or amount of work possible. / tolerance: The permitted variation from a specified value or dimension. / tradeoff: A choice that improves one objective while giving up something on another.

Speaking task: Prepare for two minutes. Speak for 45-60 seconds using the case facts, then respond to your partner's question. Switch roles and repeat without reading the model response.

Partner: You need to know exactly what information is missing. Give one answer only after your partner asks a focused question; then ask them to confirm their understanding.

Writing task: Write a 70-110 word message for the person who needs to act on this case. State the purpose, preserve the relevant facts, and make the next action or unresolved question clear. Use a subject line and an appropriate opening. Do not invent a deadline, finding, or approval.

Optional second-round challenge: The other person uses an ambiguous word such as 'ready', 'approved', or 'urgent'. Clarify it before continuing.

Model for comparison AFTER my attempt, not a response to give me first: Could you define the target weight, operating conditions, and expected life? We need measurable requirements to discuss the tradeoffs and evaluate a design consistently.

END REFERENCE

TASK: Grammar that changes the meaning

1. For a lesson, use its language workshop as the focus. For a dialogue, quote one actual sentence and select one useful grammar feature from it, such as question word order, tense, modality, conditionals or clause linking. Explain in no more than 90 words how this feature helps the speakers do their work. Keep the grammar target narrow.
2. Give two short contrasting sentences using this workplace context. Explain the difference in time, certainty, condition or politeness. Label invented examples and preserve the distinction between possible, planned and confirmed events. Describe context-dependent choices as choices, not universal rules.
3. Give me one editing or sentence-building task using a known case fact. Do not copy the supplied editing example or reveal its answer. Ask me to explain my intended meaning. Stop and wait.
4. After I answer, quote my wording, identify at most two issues, and give a brief explanation and one hint. Ask me to revise before offering a full corrected version. Then accept any accurate, natural alternative that serves the purpose.
5. Continue with two new tasks, one at a time: first guided, then an original response without a sentence frame. End with a two-sentence workplace message using the target feature and a compact self-check. Do not replace this sequence with a worksheet and answer key.

END PROMPT

Copy this prompt online or choose another lesson and support level